

ELECTRONIC DEVICES & CIRCUITS

COMPLEX ENGINEERING PROBLEM

Semester 2 — Group 3



Instructors

Dr. Mohsin Riaz

Dr. Muhammed Aqil

Design and Analysis of an AC to DC Power Supply

using Bridge Rectifier and Zener Voltage Regulation

Group Members

Name	Registration Number
Ali Ahmad Nasir	02-3-1-007-2025
Muhammed Sufyan	02-3-1-055-2025
Fatima Malik	02-3-1-021-2025

Pakistan Institute of Engineering and Applied Sciences

Date: 13 May 2026

Table of Contents

Contents

Abstract	1
1 Introduction	1
2 Objectives	1
2.1 Given Parameters	2
3 Circuit Diagram	3
3.1 Original Circuit Schematic	3
4 Input AC Signal	3
5 Bridge Rectification	4
6 Positive and Negative Half Cycle Operation	4
7 Ripple Reduction Using Capacitor	4
8 Results: Maximum and Minimum Voltages	6
8.1 Maximum Voltage Across Load	6
8.2 Minimum Voltage Across Load	7
9 Series Resistor R_s	7
10 Voltage Regulation	9
11 Final Output Across Load	10
12 Ripple Voltage Analysis	10
13 Load and Line Regulation	11
13.1 Line Regulation	11
13.2 Load Regulation	12

14 Discussion	13
15 Conclusion	14
16 References	14

Figures

List of Figures

1	Flow Chart of Power Supply Design	2
2	Complete AC to DC Power Supply Circuit (LTspice)	3
3	Complete AC to DC Power Supply Circuit Schematic	3
4	Input AC Signal Waveform	3
5	Bridge Rectifier Output Waveform	4
6	Positive and Negative Half Cycle Conduction	4
7	Ripple Reduction with Capacitor Filter	4
8	Full Wave Rectified Output with Filtering	6
9	Maximum Output Voltage (Bridge Rectifier Output)	6
10	Maximum Output Voltage from Measurement Tools	7
11	Minimum Output Voltage (Bridge Rectifier Output)	7
12	Minimum Output Voltage from Measurement Tools	7
13	Voltage Regulation Using Zener Diode	9
14	Final Regulated Output Voltage Across $1k\Omega$ Load (5.9V)	10

Abstract

Power supply circuits are fundamental components of almost every electronic system. Most electronic devices operate on direct current (DC), whereas the electrical energy supplied from utility power systems is in the form of alternating current (AC). Therefore, efficient conversion of AC power into stable DC voltage is essential for reliable operation of electronic equipment.

This project presents the design and analysis of an AC to DC power supply using a bridge rectifier, capacitor filter, and Zener diode voltage regulator. The AC input waveform is first converted into pulsating DC through full-wave bridge rectification. A filter capacitor is then used to smooth the waveform and reduce ripple components. Finally, a Zener diode operating in reverse breakdown region provides voltage regulation and stabilizes the output voltage across the load.

The final output voltage measured across a $1k\Omega$ load resistor was approximately $5.9V$, demonstrating effective rectification, filtering, and voltage regulation.

1 Introduction

Electronic circuits require a stable and constant DC voltage source for proper operation. However, the electrical energy available from power distribution systems is alternating current. As a result, electronic systems require power conversion circuits that transform AC voltage into regulated DC voltage.

An AC to DC power supply typically consists of three main stages:

- ▶ Rectification
- ▶ Filtering
- ▶ Voltage Regulation

2 Objectives

- ▶ To design and simulate an AC to DC power supply circuit.
- ▶ To implement full-wave rectification using a bridge rectifier configuration.
- ▶ To analyze ripple formation and reduce ripple voltage.

- ▶ To apply Zener diode voltage regulation.
- ▶ To measure the final DC output across a resistive load.

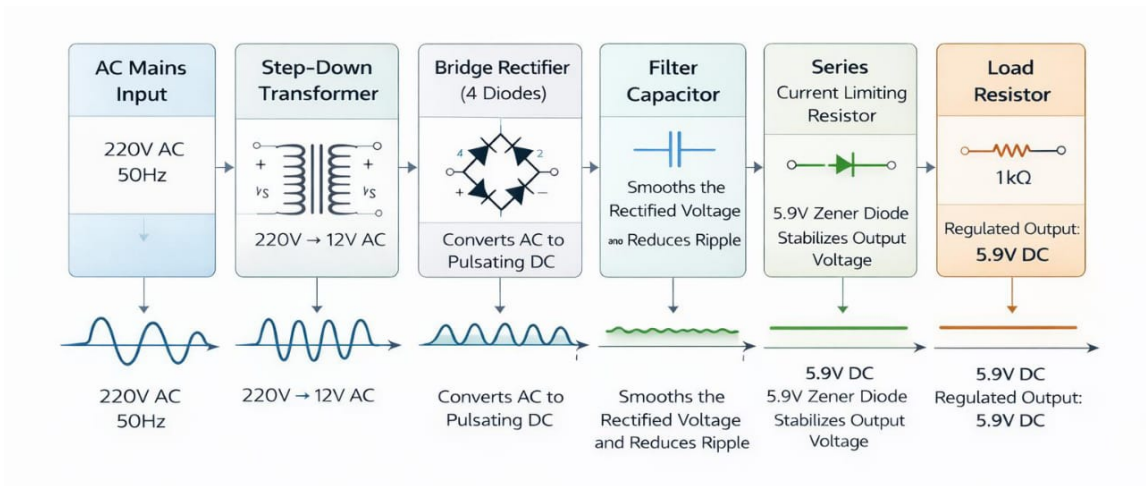


Figure 1: Flow Chart of Power Supply Design

2.1 Given Parameters

Parameter	Value
Output Voltage	$V_{out} = 5.9V$
Load Resistance	$R_L = 1k\Omega$
Transformer Secondary RMS Voltage	12V
Bridge Rectifier Drop	1.4V
Zener Voltage	5.2V
Series Diode Drop	0.7V
Mains Frequency	50Hz
Ripple Frequency (Full Wave)	100Hz

Table 1: Design Parameters

3 Circuit Diagram

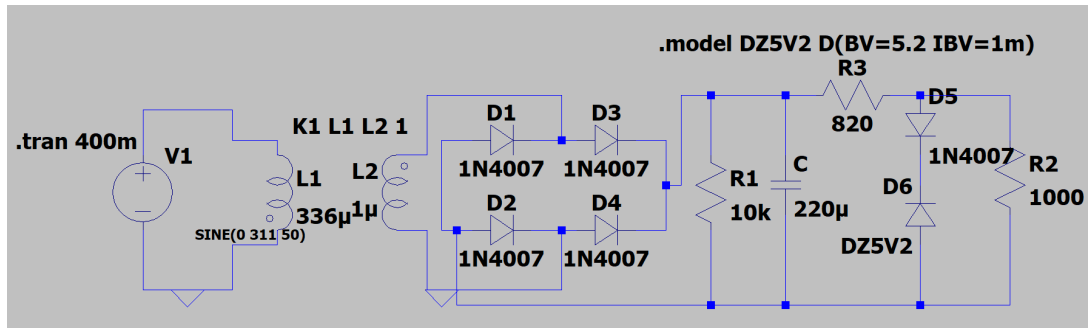


Figure 2: Complete AC to DC Power Supply Circuit (LTspice)

3.1 Original Circuit Schematic

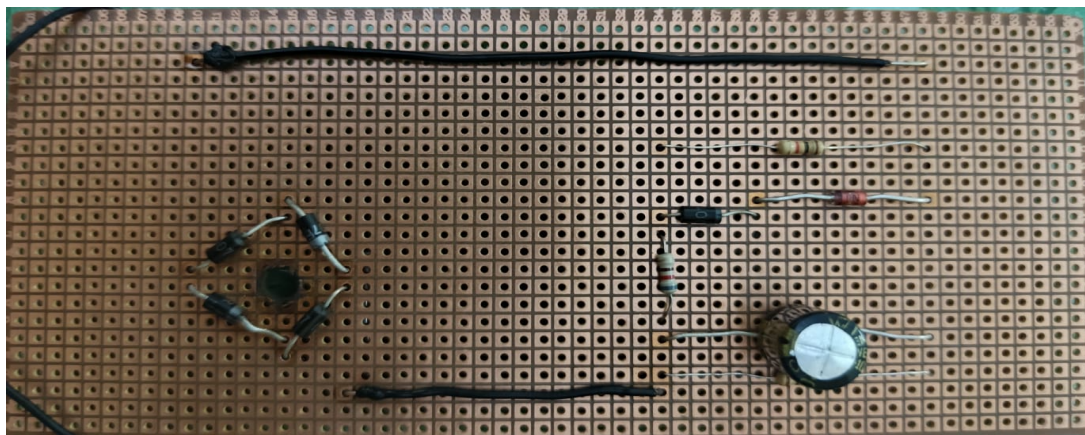


Figure 3: Complete AC to DC Power Supply Circuit Schematic

4 Input AC Signal

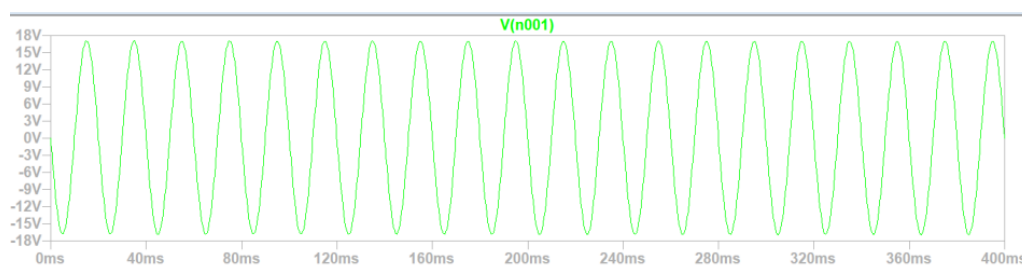


Figure 4: Input AC Signal Waveform

5 Bridge Rectification

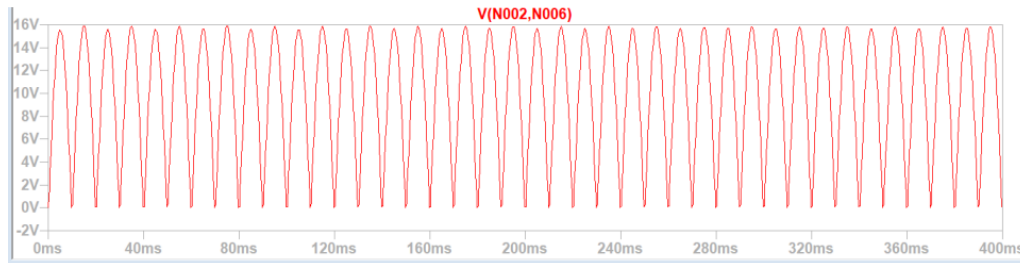


Figure 5: Bridge Rectifier Output Waveform

$$V_{peak} = V_{rms} \sqrt{2}$$

$$V_{peak} = 12 \times 1.414 = 16.97V$$

$$V_{peak(rect)} = 16.97 - 1.4$$

$$V_{peak(rect)} \approx 15.6V$$

6 Positive and Negative Half Cycle Operation

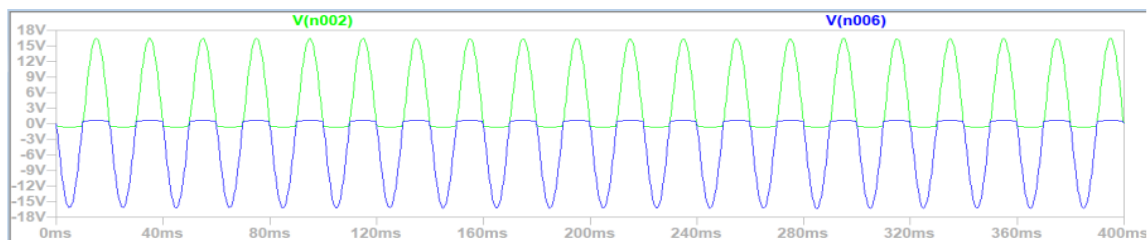


Figure 6: Positive and Negative Half Cycle Conduction

7 Ripple Reduction Using Capacitor

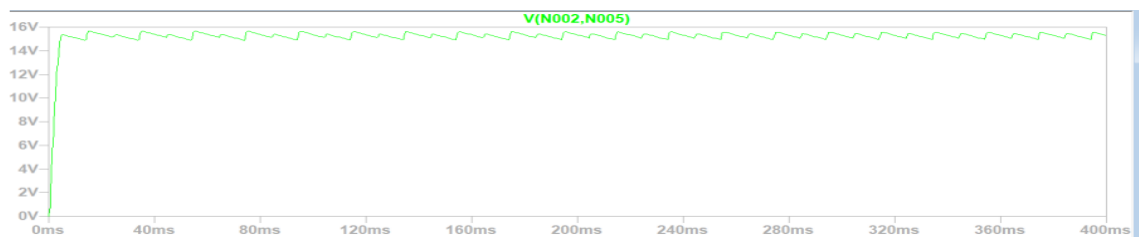


Figure 7: Ripple Reduction with Capacitor Filter

For a full-wave rectifier, the ripple voltage is given by

$$V_r = \frac{I_L}{fC}$$

Rearranging the equation to calculate the capacitor value:

$$C = \frac{I_L}{fV_r}$$

Ripple Voltage Estimation

The ripple voltage is assumed to be approximately 5% of the output voltage:

$$V_r \approx 5\% \times V_{out}$$

$$V_r = 0.05 \times 5.9$$

$$V_r = 0.295 \text{ V}$$

Given Values

- ▶ Load current: $I_L = 0.0059 \text{ A}$
- ▶ Frequency: $f = 2 \times 50 \text{ Hz} = 100 \text{ Hz}$

Capacitor Calculation

$$C = \frac{0.0059}{100 \times 0.295}$$

$$C = \frac{0.0059}{29.5}$$

$$C = 200 \mu\text{F}$$

Ripple Verification

Using the selected capacitor value $C = 220 \mu\text{F}$:

$$V_r = \frac{0.0059}{100 \times 220 \times 10^{-6}}$$

$$V_r = \frac{0.0059}{0.022}$$

$$V_r \approx 0.268 \text{ V}$$

Selected Capacitor Value: $C = 220 \mu\text{F}$



Figure 8: Full Wave Rectified Output with Filtering

8 Results: Maximum and Minimum Voltages

8.1 Maximum Voltage Across Load

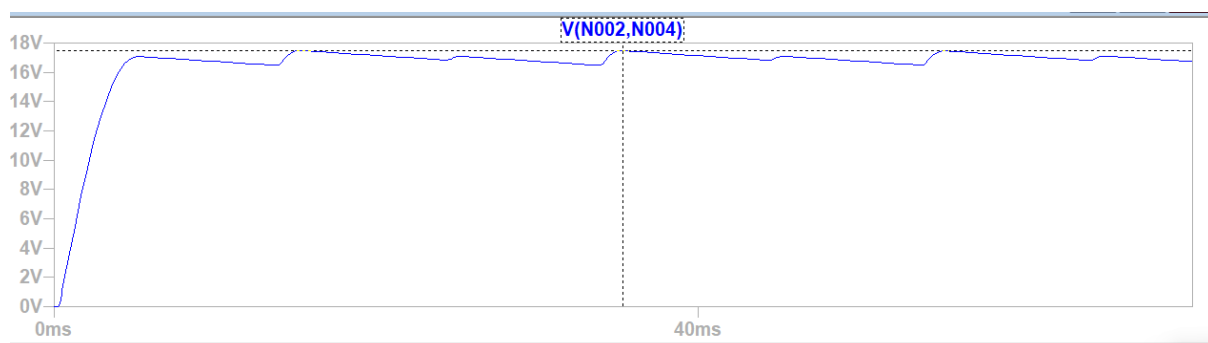


Figure 9: Maximum Output Voltage (Bridge Rectifier Output)

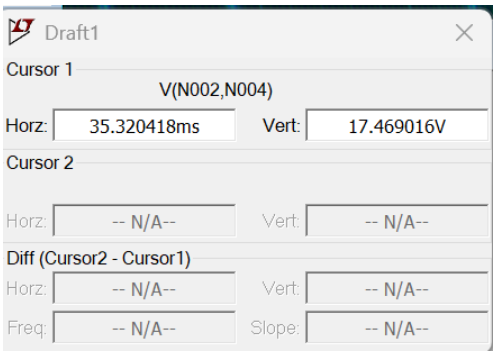


Figure 10: Maximum Output Voltage from Measurement Tools

8.2 Minimum Voltage Across Load

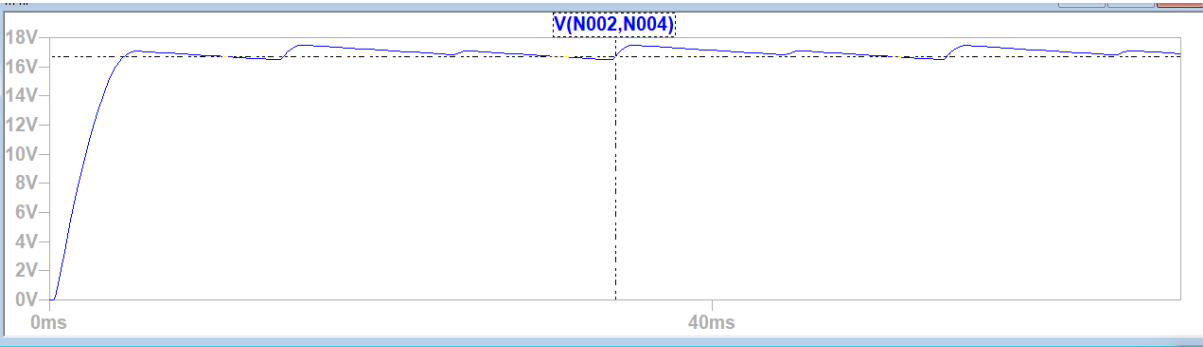


Figure 11: Minimum Output Voltage (Bridge Rectifier Output)

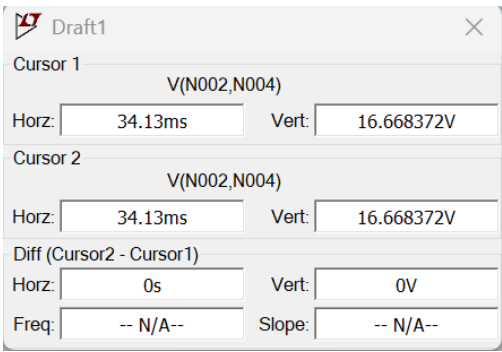


Figure 12: Minimum Output Voltage from Measurement Tools

9 Series Resistor R_s

Formula

$$R_s = \frac{V_{in,peak} - V_{out}}{I_L + I_Z}$$

Where

- Voltage across resistor:

$$V_R = V_{in,peak} - V_{out}$$

$$V_R = 15.6 - 5.9 = 9.7 \text{ V}$$

- Load current:

$$I_L = 5.9 \text{ mA} = 0.0059 \text{ A}$$

- Chosen Zener current:

$$I_Z = 5 \text{ mA} = 0.005 \text{ A}$$

Calculation

$$R_s = \frac{9.7}{0.0059 + 0.005}$$

$$R_s = \frac{9.7}{0.0109}$$

$$R_s \approx 890.8 \Omega$$

Selected Standard Value

$$R_s = 820 \Omega$$

Series Current

$$I_R = \frac{V_R}{R_s}$$

$$I_R = \frac{9.7}{820}$$

$$I_R \approx 11.83 \text{ mA}$$

Zener Current

$$I_Z = I_R - I_L$$

$$I_Z = 11.83 - 5.9$$

$$I_Z \approx 5.93 \text{ mA}$$

10 Voltage Regulation

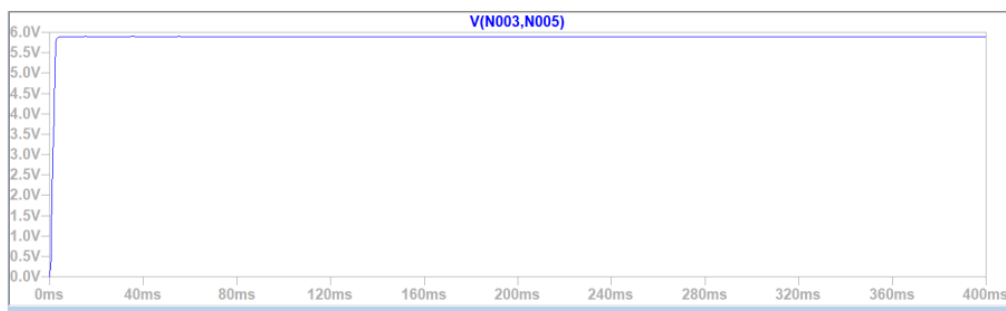


Figure 13: Voltage Regulation Using Zener Diode

- Zener voltage:

$$V_Z = 5.2 \text{ V}$$

- Series diode drop:

$$V_D = 0.7 \text{ V}$$

Output Voltage

$$V_{out} = V_Z + V_D = 5.2 + 0.7$$

$$V_{out} = 5.9 \text{ V}$$

Load Current

$$I_L = \frac{V_{out}}{R_L}$$

$$I_L = \frac{5.9}{1000}$$

$$I_L = 5.9 \text{ mA}$$

11 Final Output Across Load

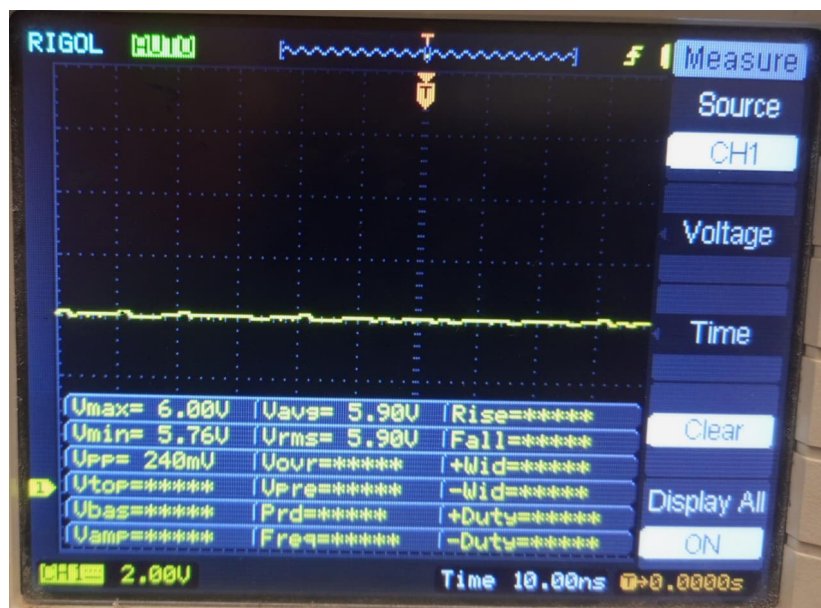


Figure 14: Final Regulated Output Voltage Across $1k\Omega$ Load (5.9V)

$$V_{out} \approx 5.9V$$

12 Ripple Voltage Analysis

For an output voltage of 5.9 V, the theoretical ripple voltage was expected to be:

$$V_{r(\text{theoretical})} = 0.268 \text{ V}$$

The theoretical ripple percentage is:

$$\%r_{\text{theoretical}} = \frac{0.268}{5.9} \times 100$$

$$\%r_{\text{theoretical}} = 4.54\%$$

However, the ripple voltage measured on the oscilloscope was:

$$V_{r(\text{practical})} = 0.240 \text{ V}$$

The practical ripple percentage is:

$$\%r_{\text{practical}} = \frac{0.240}{5.9} \times 100$$

$$\%r_{\text{practical}} = 4.07\%$$

The percentage difference between theoretical and practical ripple percentages is:

$$\% \text{Error} = \frac{|4.54 - 4.07|}{4.54} \times 100$$

$$\% \text{Error} = 10.35\%$$

The variation between theoretical and practical values is due to non-ideal component behavior, measurement limitations, and practical circuit conditions.

13 Load and Line Regulation

Given Values

- ▶ Output Voltage:

$$V_{out} = 5.9 \text{ V}$$

- ▶ Series Resistor:

$$R_s = 820 \Omega$$

- ▶ Zener Dynamic Resistance:

$$r_z = 30 \Omega$$

13.1 Line Regulation

Line regulation represents the change in output voltage due to variation in input voltage.

$$\text{Line Regulation} = \frac{\Delta V_{out}}{\Delta V_{in}}$$

For a Zener regulator:

$$\Delta V_{out} = \Delta V_{in} \left(\frac{r_z}{R_s + r_z} \right)$$

Assuming an input variation of:

$$\Delta V_{in} = \pm 1 \text{ V}$$

$$\Delta V_{out} = 1 \times \left(\frac{30}{820 + 30} \right)$$

$$\Delta V_{out} = \frac{30}{850}$$

$$\Delta V_{out} \approx 0.0353 \text{ V}$$

$$\Delta V_{out} \approx 35.3 \text{ mV}$$

Therefore,

$$\text{Line Regulation} \approx 35.3 \text{ mV/V}$$

13.2 Load Regulation

Load regulation represents the change in output voltage from no-load to full-load condition.

Using the approximation:

$$\text{Load Regulation} = I_L \times r_z$$

$$\text{Load Regulation} = (5.9 \times 10^{-3}) \times 30$$

$$\text{Load Regulation} = 0.177 \text{ V}$$

$$\text{Load Regulation} = 177\text{ mV}$$

Percentage load regulation is:

$$\% \text{Load Regulation} = \frac{0.177}{5.9} \times 100$$

$$\% \text{Load Regulation} \approx 3\%$$

14 Discussion

The simulation and analysis of the designed circuit confirm that it effectively performs AC to DC conversion with satisfactory performance. The system consists of a bridge rectifier, capacitor filter, series resistor, and Zener diode regulator, which together ensure proper rectification, smoothing, and voltage stabilization of the output.

The bridge rectifier converts both the positive and negative halves of the AC input into a pulsating DC waveform, making better use of the input signal compared to a half-wave rectifier. Due to full-wave rectification, the ripple frequency is doubled to 100 Hz , and after diode drops, the peak rectified voltage becomes approximately 15.6 V .

After rectification, the capacitor filter is used to reduce ripple in the output. The $220\text{ }\mu\text{F}$ capacitor charges near the peak voltage and discharges slowly through the load when the input voltage falls, producing a smoother DC waveform. The theoretical ripple voltage was calculated as 0.268 V (about 4.54%), while the oscilloscope reading showed approximately 0.240 V (about 4.07%), which is reasonably close and indicates effective filtering.

The Zener diode regulator further stabilizes the output voltage by operating in the breakdown region. With the combination of a 5.2 V Zener diode and a 0.7 V diode drop, the regulated output voltage is maintained at approximately 5.9 V across the $1\text{ k}\Omega$ load. The series resistor ensures proper current limiting and helps maintain stable Zener operation under varying conditions.

Overall, the results show good agreement between theoretical calculations, simulation, and practical observations. Small variations in ripple and voltage values are due to non-ideal components and real circuit behavior. The designed circuit provides a stable, low-ripple DC output and is suitable for basic low-power electronic applications.

15 Conclusion

In this project, an AC to DC power supply circuit was successfully designed, simulated, and analyzed using fundamental electronic components, including a bridge rectifier, capacitor filter, and Zener diode voltage regulator. The complete system effectively converts an alternating input signal into a stable and regulated direct current output.

The measured output voltage across the $1k\Omega$ load resistor was approximately 5.9 V , which closely matches the expected regulated value obtained from theoretical calculations. This confirms that the circuit is working properly in terms of rectification, filtering, and voltage regulation. The small difference between ideal and practical values is acceptable and reflects normal circuit behavior.

Overall, the results clearly demonstrate the practical application of basic electronic components in the design of reliable DC power supplies. The combination of bridge rectification, capacitor smoothing, and Zener regulation provides a stable output suitable for low-power electronic devices.

This project helps strengthen the understanding of core concepts such as AC to DC conversion, ripple reduction, and voltage regulation, while also giving useful insight into real-world circuit performance. The designed system is simple, economical, and effective, making it a good foundation for more advanced power supply designs in future work.

16 References

1. Sedra, A. S., Smith, K. C., *Microelectronic Circuits*, 8th Edition, Oxford University Press.
2. Boylestad, R., Nashelsky, L., *Electronic Devices and Circuit Theory*, 11th Edition, Pearson.
3. Basic Electrical Engineering Laboratory Manual, PIEAS.